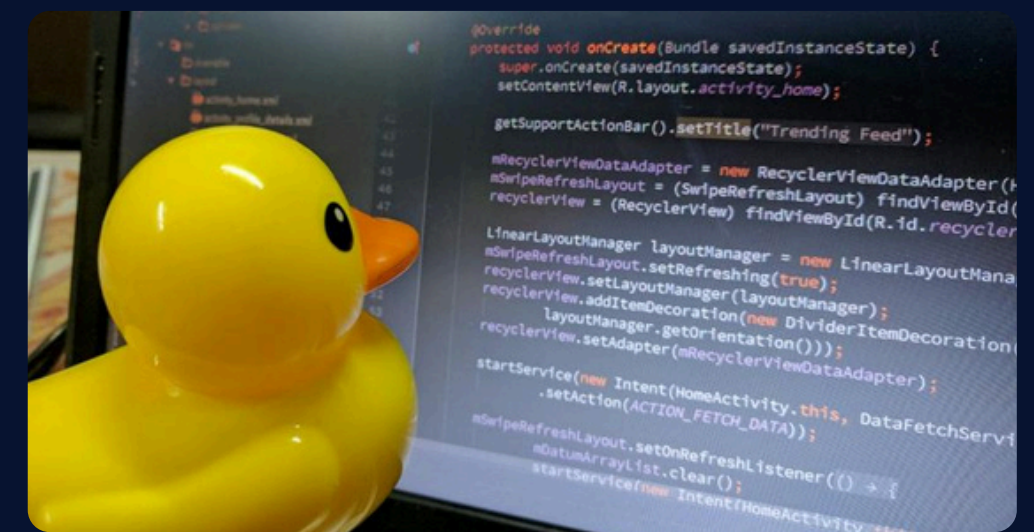




Optimization & Evaluation

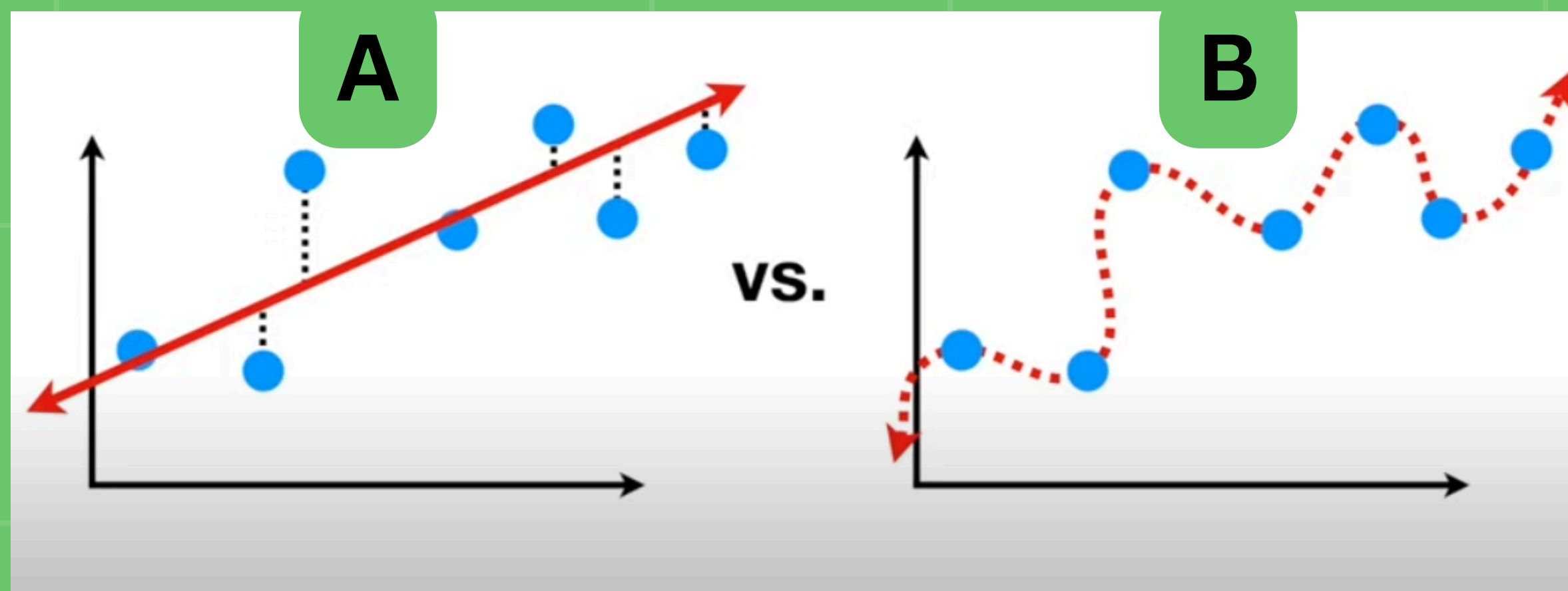


Session -> 6



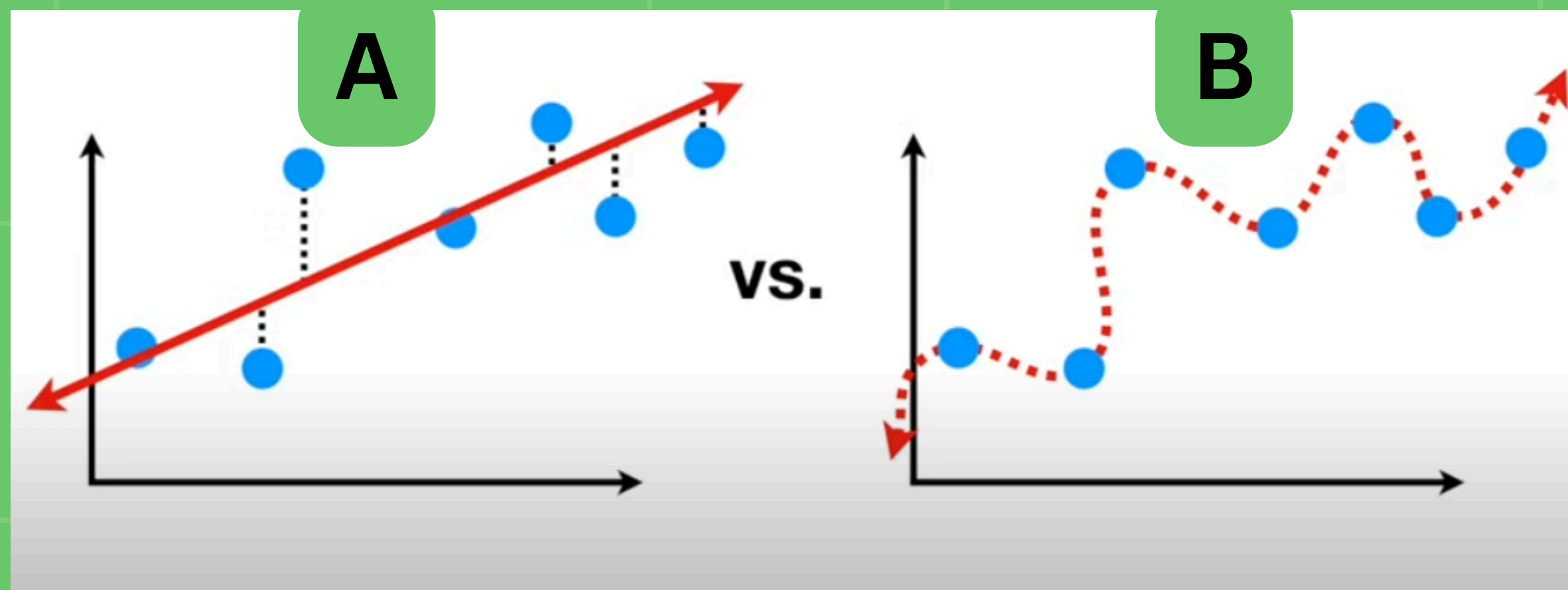
Bias & Variance

Bias **vs** Variance



If we have a training dataset plotted as these blue points, we can see that the model(B) can predict all of them correctly, while model (A) is close to them but still the cost is greater than 0

Bias **vs** Variance

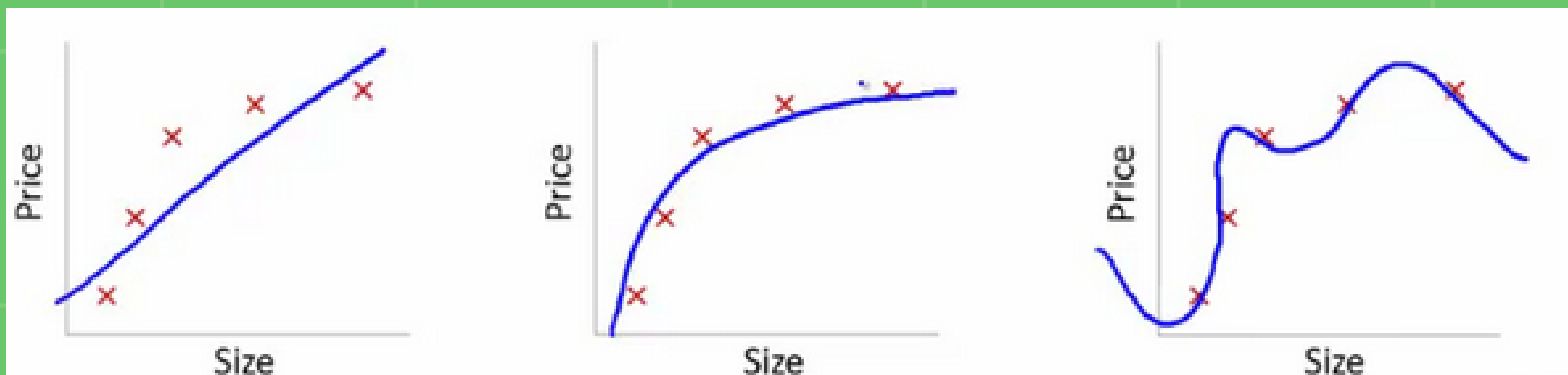


In this example, we can say that model(A) is more biased than mode(B)

Bias **vs** Variance

What does bias mean?

- When there is an error, there is bias; when the error is large, there is a high bias.
- if the model is highly biased, that means the algorithm is too simple to catch the patterns of the data.



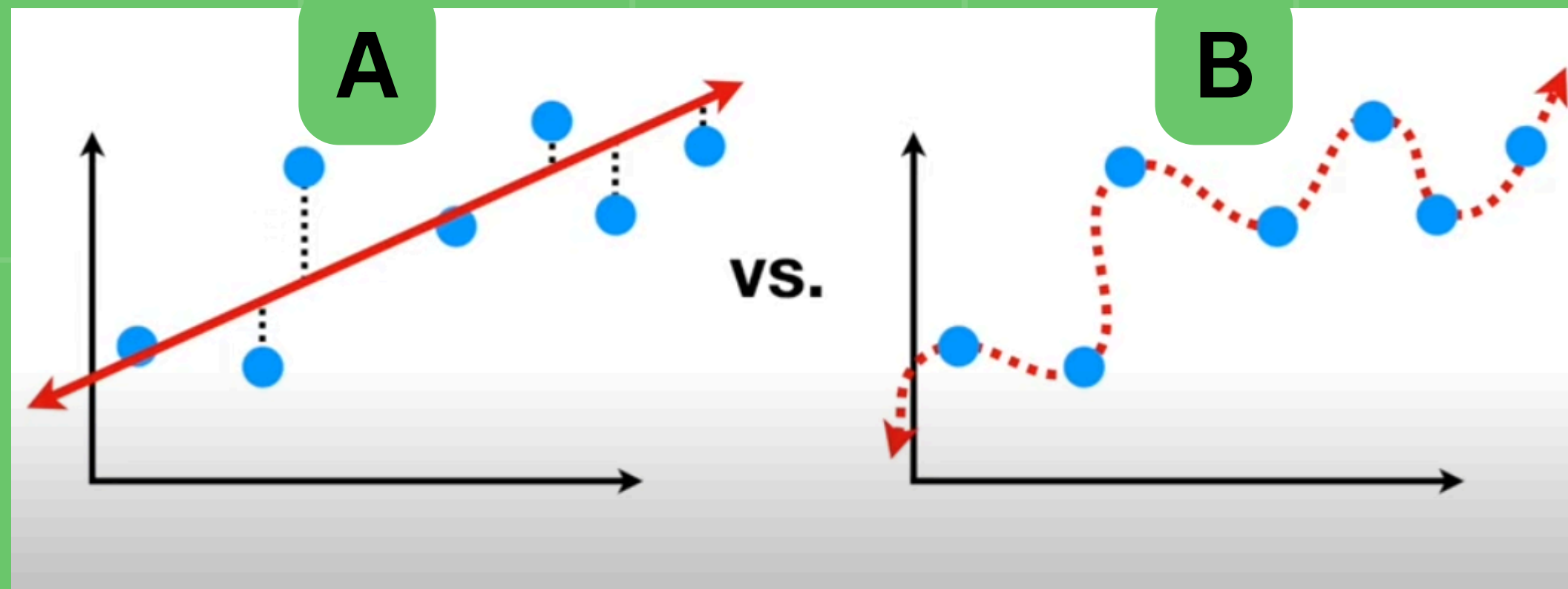
High bias

balanced

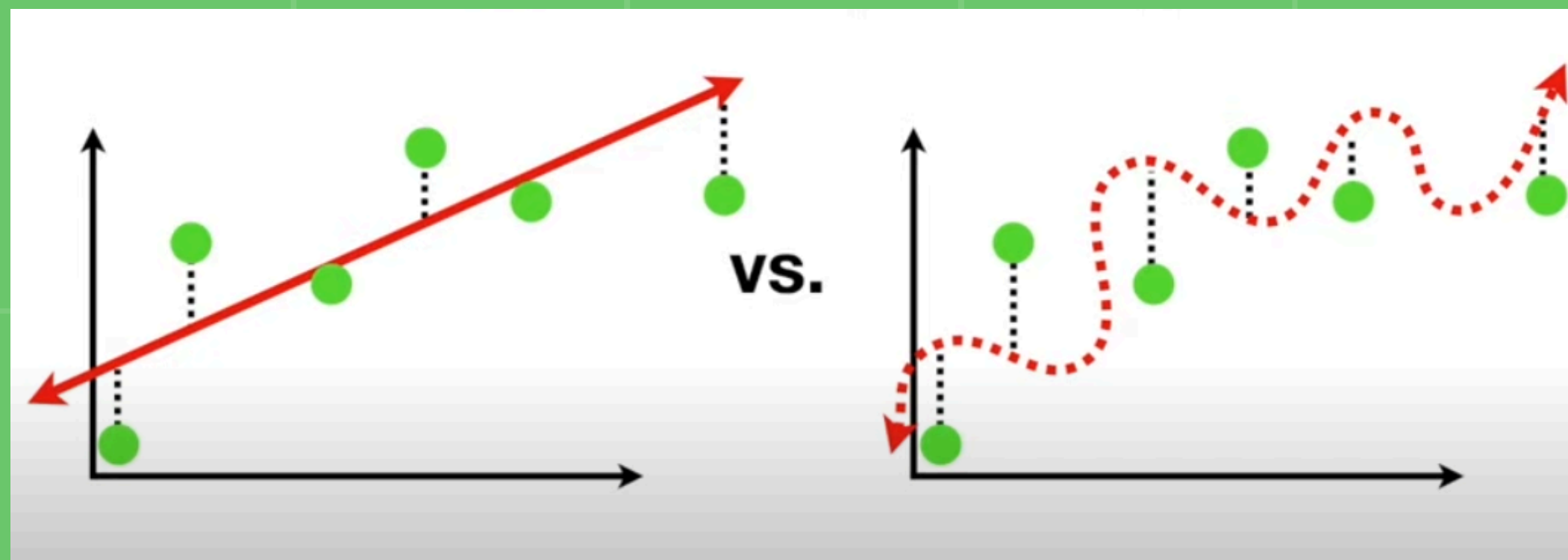
Low Bias

Bias **vs** Variance

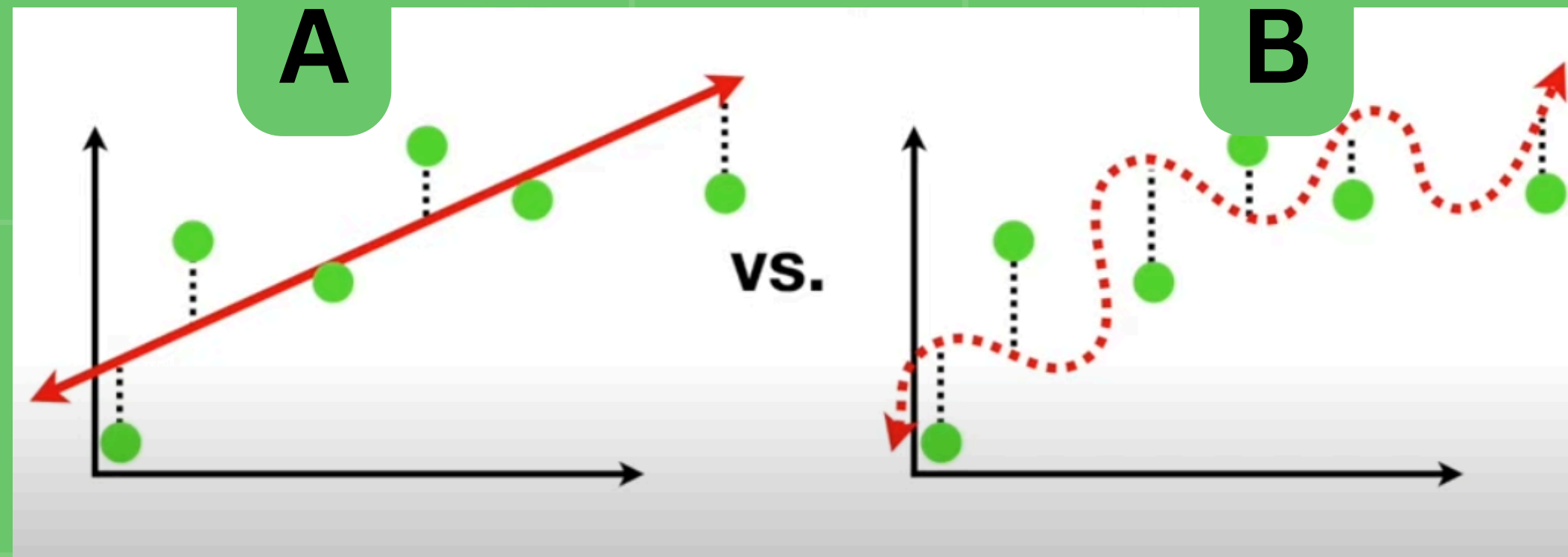
Training



Testing

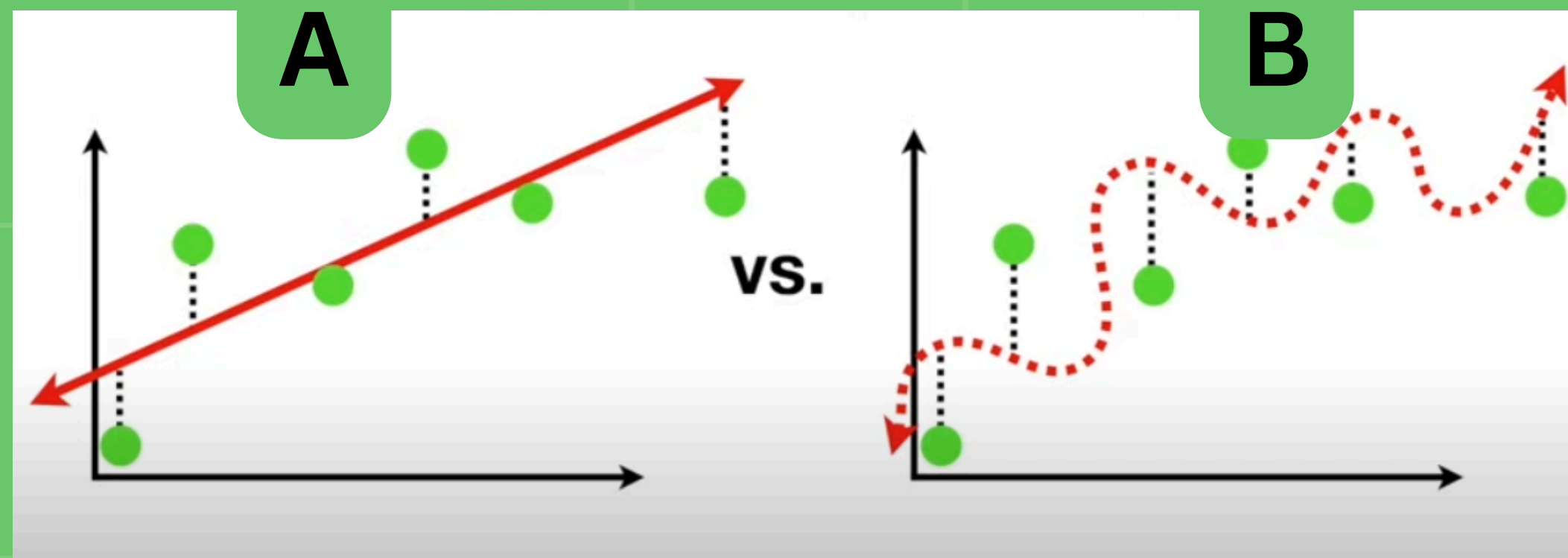


Bias **vs** Variance



In the previous image, model(B) performed better in training **but** when we plot the test data as green points, it shows that model(B) missed all the points, and not even close, unlike model(A) that performed better.

Bias **vs** Variance

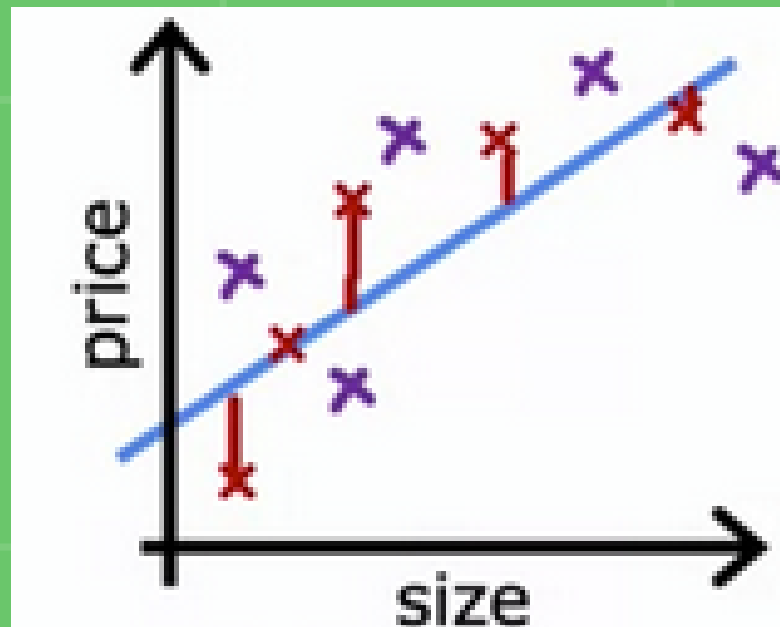


In this example, model(B) is high variance and model(A) is low variance

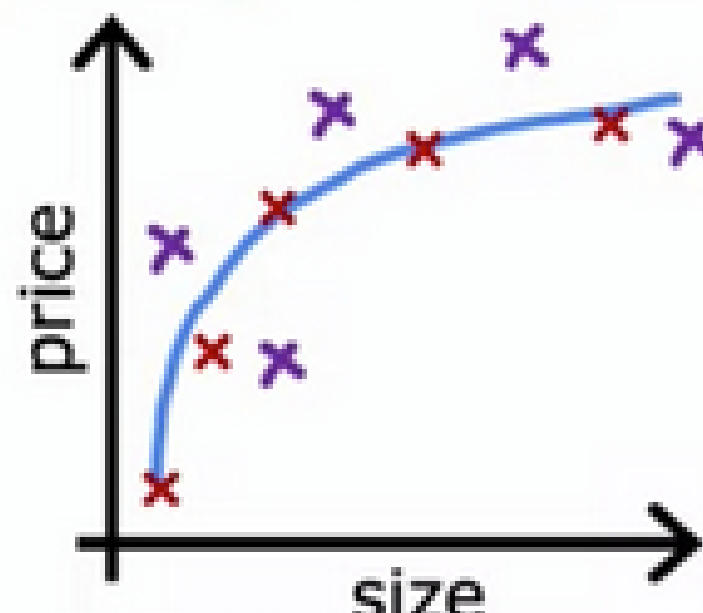
Bias **vs** Variance

What does variance mean?

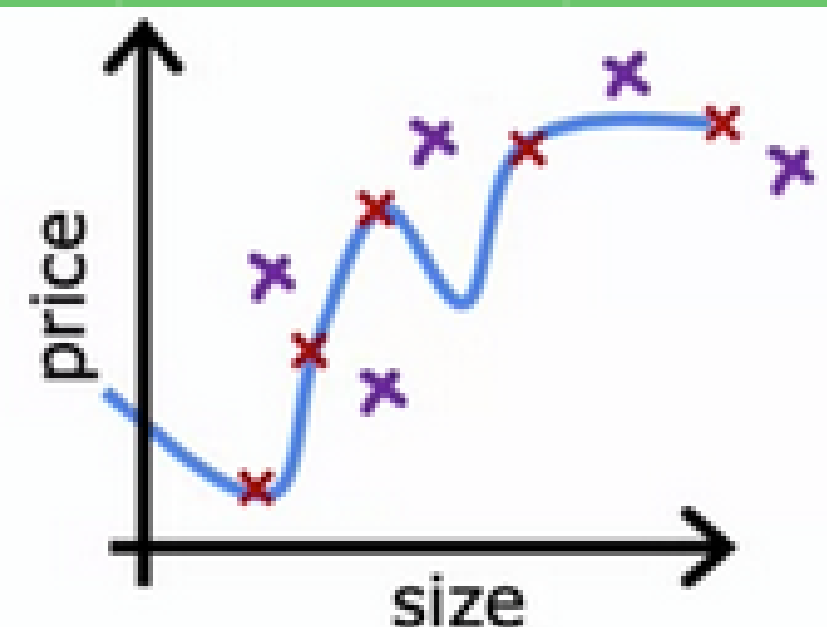
- The difference of scores between the train and test set is called 'variance', so when a model is high variance, that means it performs well in the training but poorly in the testing.



Low Variance



balanced



High Variance

“An ideal algorithm has a **low bias**
(accurately models the true
relationship), and **low variance**
(consistent predictions across
different datasets)”



Overfitting & Underfitting

Three Students and an Exam

Student 1:

- This student barely studies and doesn't understand the material well. They miss the main concepts and lack the basic understanding needed. As a result, **they get both simple and complex questions wrong** in the exam.

Three Students and an Exam

Student 2:

- This student memorizes every single practice question and answer word-for-word. They know all the textbook examples perfectly but don't understand the underlying ideas.
- On the exam, they can answer **only the repeated (memorized) questions correctly**, but fail to answer new or slightly different questions.

Three Students and an Exam

Student 3:

- This student studies the key concepts and understands the patterns in the types of questions.
- They practice with different kinds of problems and learn how to apply their knowledge.
- In the exam, they **handle both familiar** and unfamiliar questions effectively.

Three Students and an Exam

Student 1 → underfitting

Student 2 → overfitting

Student 3 → Balanced (Good Generalization)

What is Underfitting?

- **Definition:** Underfitting happens when a model is too simple to capture the underlying structure of the data.
- **Behavior:**  Train score: Low
 Test score: Low
- **Reason:** The model hasn't learned enough – it has high bias and can't capture the patterns in the data.
- **Example:** Using a linear model to fit complex, nonlinear data.

What is Overfitting?

- **Definition:** Overfitting occurs when a model is too complex and learns both the patterns and the noise in the training data.
- **Behavior:**  Train score: high
 Test score: Low
- **Reason:** it memorizes instead of generalizing.
- **Example:** A deep decision tree that perfectly fits training points, even outliers.

What is Good Generalization?

- **Definition:** A well-fit model captures the true patterns in the training data and can generalize well to new, unseen data.
- **Behavior:**  Train score: high
 Test score: high
- **Reason:** The model has the right complexity.



Regression Evaluation

Common Regression Evaluation Metrics

- 1- MAE (Mean Absolute Error)
- 2- MSE (Mean Squared Error)
- 3- R^2 Score

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

R^2 Score

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$



Classification Evaluation

Classification Evaluation Metrics

1- Accuracy

2- Recall

3- Precision

4- F1 Score

Accuracy

- Accuracy is a measure of how often the model is correct overall.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Accuracy Can be misleading

- Imagine that we have a dataset that contains data about people who have coronavirus. The dataset contains 1000 patients distributed like this:
 - 990 positive
 - 10 negative
- If we designed an algorithm that just prints negative, we would get an accuracy of 99%, but actually our solution does not work

When to use accuracy

- If the dataset is balanced (both classes have the same or close frequencies) use accuracy ✓
- If the dataset is imbalanced, use other metrics such as **recall**, **precision**, and **F1-score**

Precision

- Precision tells us out of all the times we predicted Positive, how many were actually correct.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- TP = True Positive
- FP = False Positive

Example: you predicted that 10 people are positive but only 7 are then the precision = $7 / 10 = 0.7$

Recall

- Recall tells us out of all actual Positive cases, how many we correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- TP = True Positive
- FN = False Negative

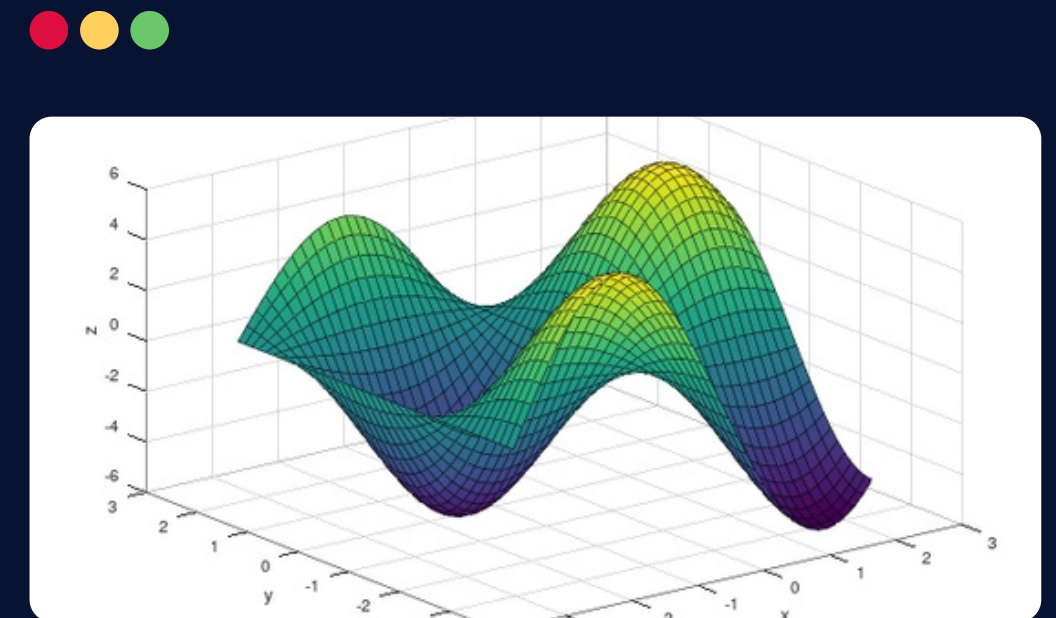
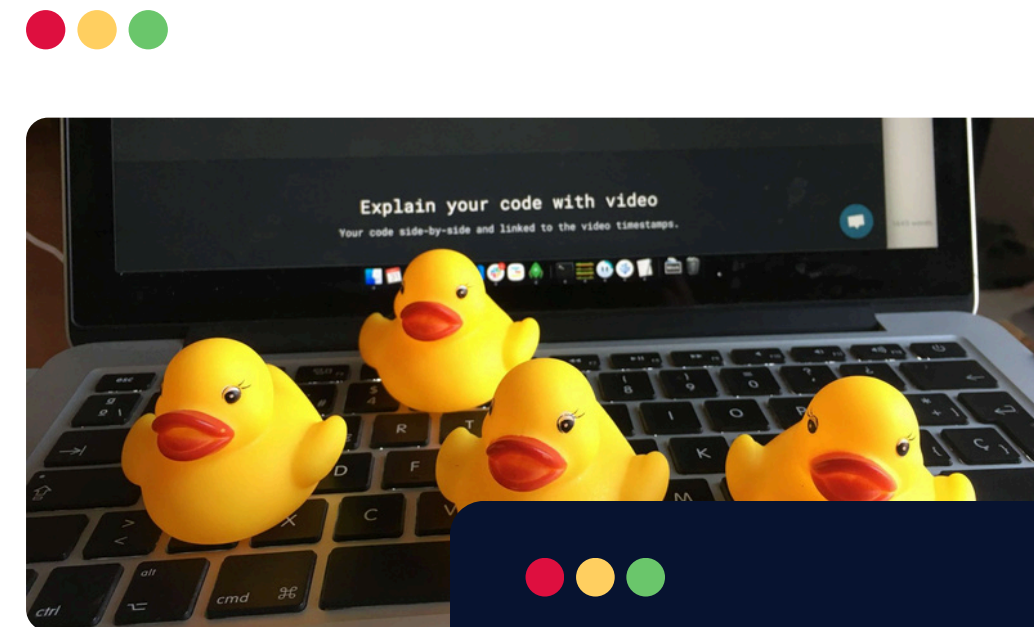
Example: The data contain 10 positive patients, and your model predicted 6 of them, $\text{recall} = 6 / 10 = 0.6$

F1 Score

- F1 Score is the harmonic mean of Precision and Recall.
- It gives you a single number that balances both.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

= If both numbers are high, the F1 score is high, if one of them is low, the F1 score will be low.



Thank You

created by:
Ahd Ehab